

Total No. of Questions : 6]

SEAT No. :

P5435

[Total No. of Pages : 2

BE/Insem./Oct.-4

B.E. (Civil)

STRUCTURAL DESIGN OF BRIDGES
(2012 Pattern) (Semester - I) (Elective - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Answer Q.1 or Q.2; Q.3 or Q.4 and Q.5 or Q.6.*
- 2) *Figures in bold to the right, indicate full marks.*
- 3) *Latest revisions of IS: 456, IRC: 6, IRC: 112 and IS: 1343 are allowed in the examination.*
- 4) *If necessary, assume suitable data and indicate clearly.*
- 5) *Use of electronic pocket calculator is allowed.*

Q1) Classify the bridges according to material of construction with suitable sketches.

[10]

OR

Q2) Explain IRC loadings for highway bridges.

[10]

Q3) What is impact loading? How is it calculated? Explain with an example. **[10]**

OR

Q4) Explain Pigeaud's Method with suitable sketch.

[10]

Q5) An R.C. T-Beam deck slab bridge for two lane National Highway has the following details. **[10]**

- a) Thickness of railings - 100 mm
- b) Thickness of footpath - 200 mm
- c) Thickness of wearing coat - 80 mm
- d) Span of main girder - 14.0 m
- e) No. of main girders - 3.5m

P.T.O.

- f) Spacing of cross-beams - 3 m c/c
- g) Live load - IRC Class A
- h) Materials - M30 grade of concrete and Fe 500 grade of steel

Adopt $m_1 = 0.06$ and $m_2 = 0.04$

Design the cantilever slab and sketch the details of reinforcement

OR

Q6) For the R.C. T-Beam deck slab bridge given in Q.4, design the interior panel and sketch the details of reinforcement. **[10]**

